POWER ELECTRONICS SYLLABUS

Power semiconductor switches and triggering devices

1. Diodes
2. BJT
3. MOSFET
4. SCR
5. IGBT
6. GTO
7. TRIAC and DIAC
8. Power computation of sinusoidal and non-sinusoidal waveforms

Uncontrolled and controlled Rectifiers

1. single phase and three-phase
2. half wave and full wave rectifiers and applications

Regulated power supplies

1. Linear dc-dc regulators
2. Choppers
3. switching dc-dc regulators: buck, buck boost, boost, Cuk, SEPIC regulators
4. DC motor control

AC voltage controllers

1. Single and three-phase AC voltage controllers
2. AC motor control
3. Cyclo-converters

Inverters

1. Full bridge inverter
2. square wave inverters
3. amplitude and harmonics control in inverters
4. Fourier series analysis
5. Multilevel inverters
6. PWM output in inverters
7. Harmonics analysis
8. three phase inverters

Resonant converters

1. Resonant switch converters: Zero current switching, zero voltage switching, series and parallel Resonant DC-DC converters

Power Electronics in Renewable Energy Systems

1. PE applications